Supplementary Information for Exact Kullback–Leibler Boundaries for Threshold-Decision Stability in Probabilistic Classifiers

Nicholas Christakis and Dimitris Drikakis

Overview

This supplementary file contains the detailed figures generated by the analysis script `generate_multidomain_stability_figures.py`. The main manuscript reports the all-threshold heatmaps and focuses the detailed progressive analysis on $\tau = 0.5$. This supplement provides the threshold-specific diagnostic scatter plots for $\tau = 0.3$, $\tau = 0.5$ and $\tau = 0.7$, together with final ROC curves for each dataset-model pair.

The threshold-certificate scatter plots include flipped points only. The horizontal axis is the conservative Pinsker certificate,

$$C_{\text{Pinsker}} = 2|p_{\text{prev}} - \tau|^2,$$

and the vertical axis is the observed Bernoulli KL displacement,

$$D_{\text{step}} = D_{\text{KL}}(\text{Bern}(p_{\text{curr}}) \| \text{Bern}(p_{\text{prev}})) \,.$$

The dashed diagonal corresponds to $D_{\text{step}} = C_{\text{Pinsker}}$. Threshold-crossing observations must lie on or above this boundary up to numerical precision.

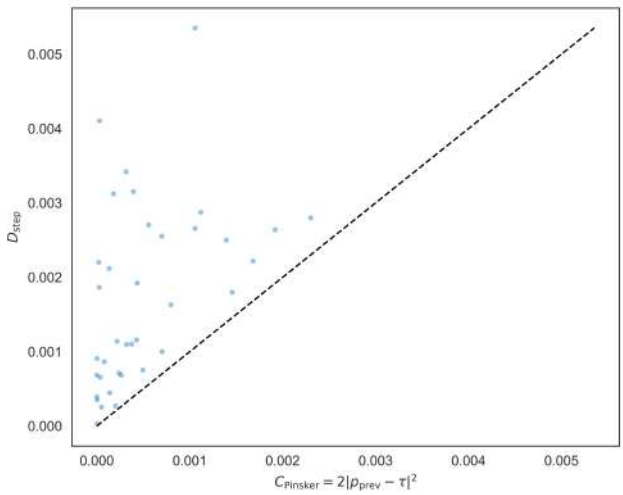
Table S1: Dataset–model–threshold combinations for which no threshold-certificate scatter plot was generated because no flipped validation instances were observed in the reported analysis run. A missing panel therefore represents an absence of observed flips rather than missing data or a failed analysis.

Dataset	Model	Threshold
Machine failure	Logistic regression	$\tau = 0.5$
Machine failure	Logistic regression	$\tau = 0.7$
Machine failure	ANN	$\tau = 0.7$

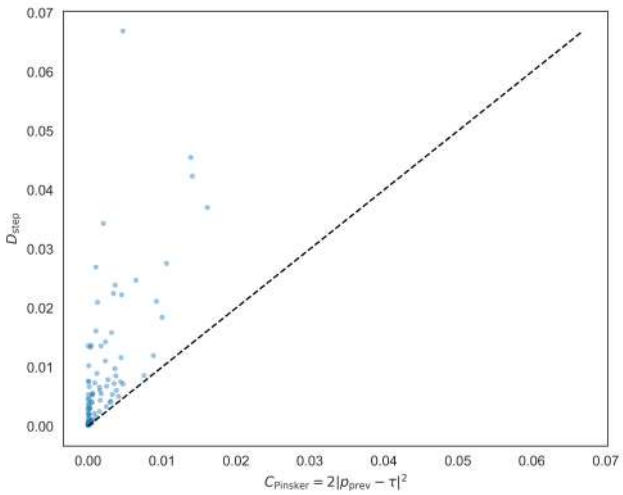
Table S1 lists the dataset–model–threshold combinations for which no threshold-certificate scatter plot was generated. In these cases, no flipped validation instances were observed in the corresponding analysis run. Accordingly, the absence of a scatter-plot panel indicates that there were no observed flips to plot, rather than missing data, an omitted analysis, or an analysis failure.

1 Threshold-certificate scatter plots

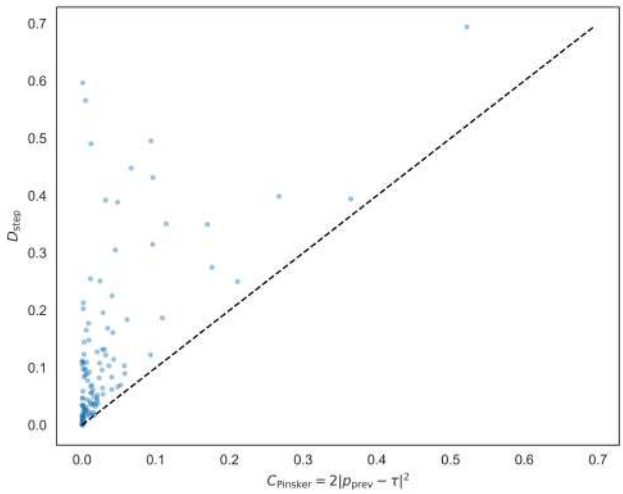
1.1 PDAC biomarkers



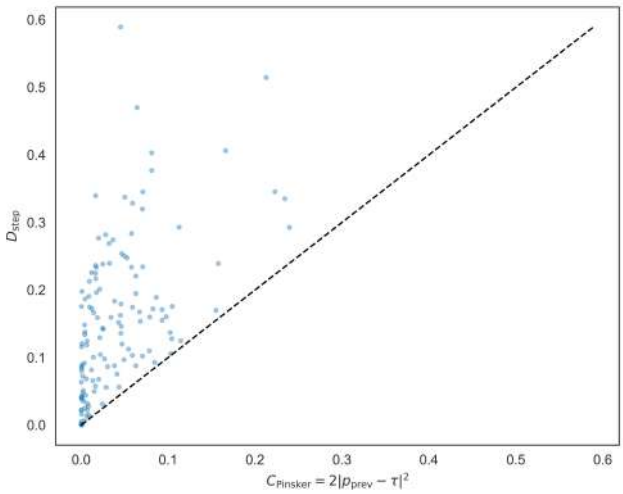
(a) Logistic regression



(b) Random forest

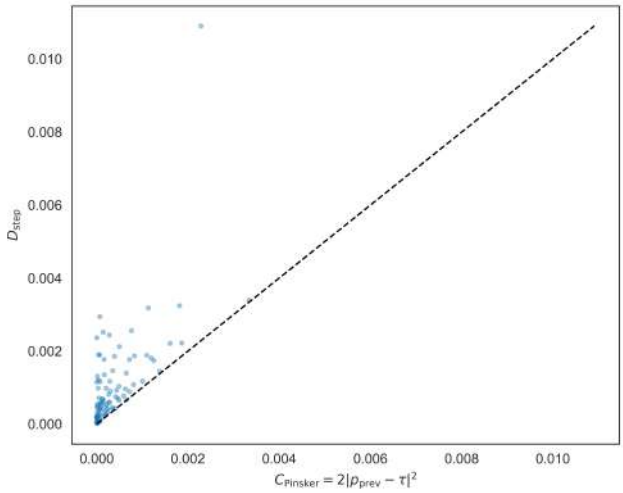


(c) XGBoost

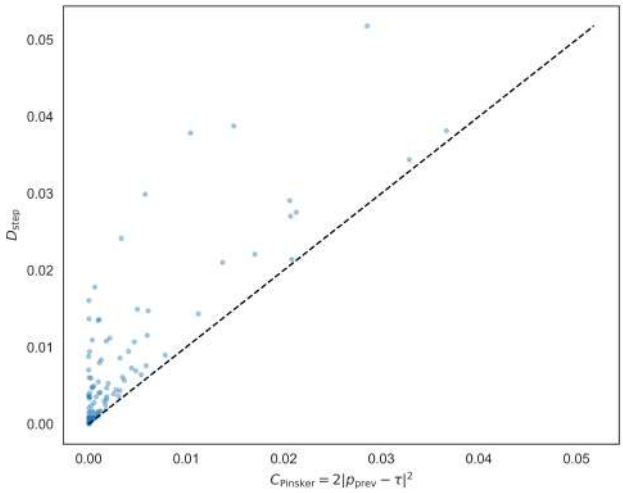


(d) ANN

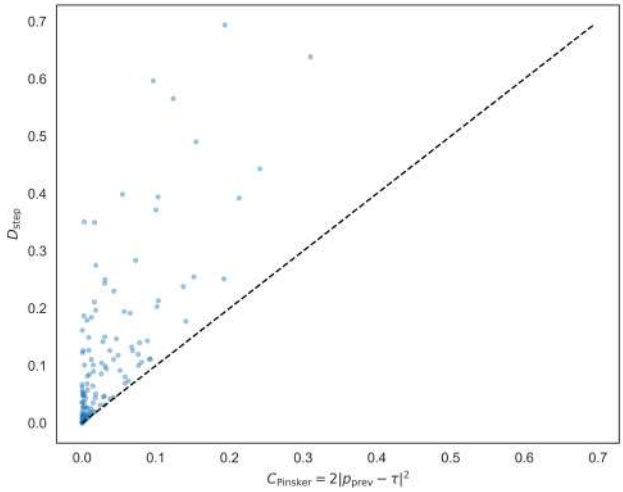
Figure S1: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.3$.



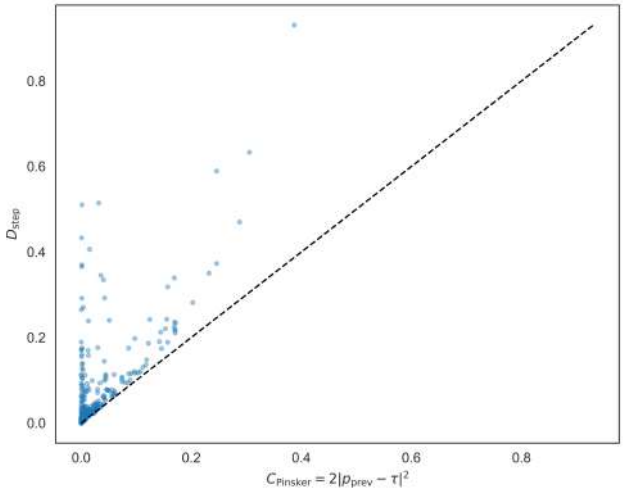
(a) Logistic regression



(b) Random forest

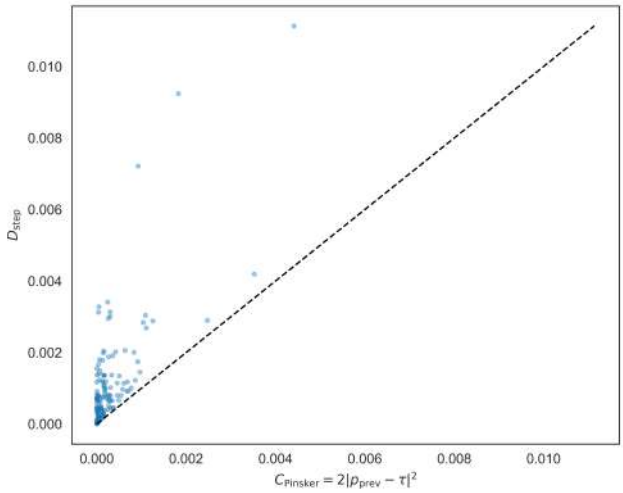


(c) XGBoost

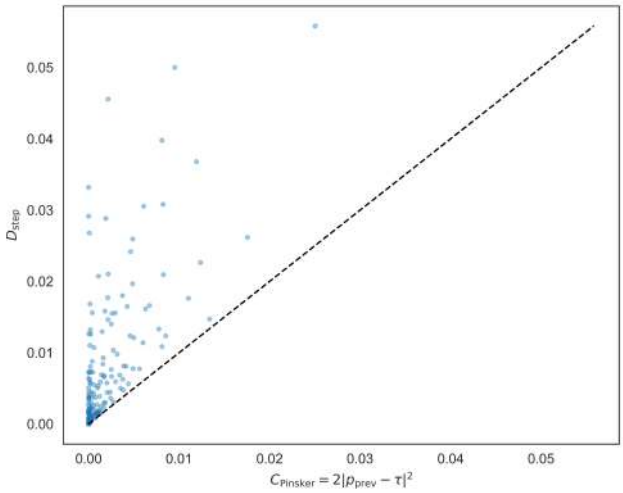


(d) ANN

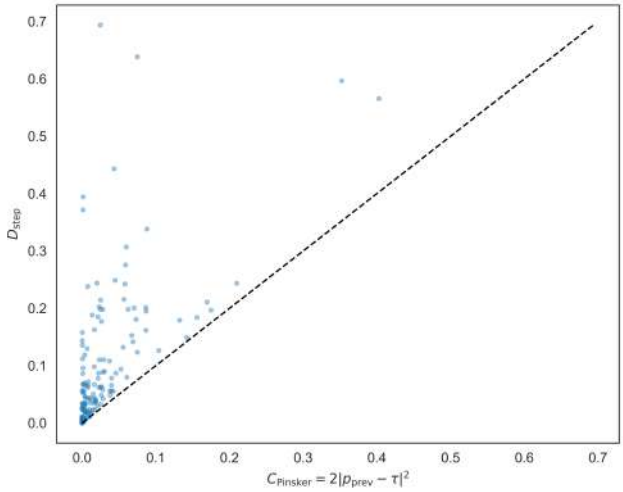
Figure S2: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.5$.



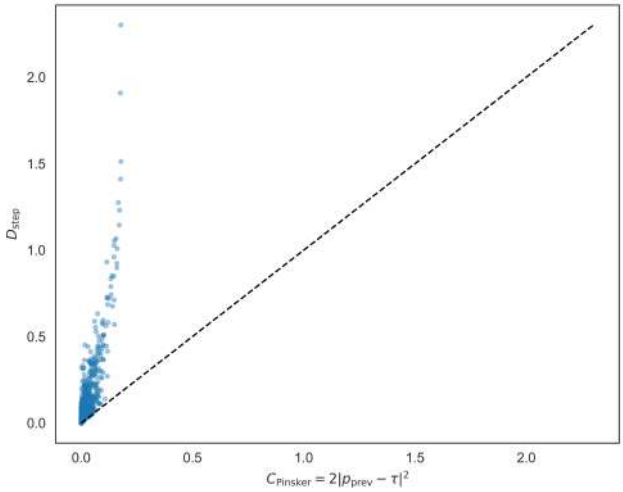
(a) Logistic regression



(b) Random forest



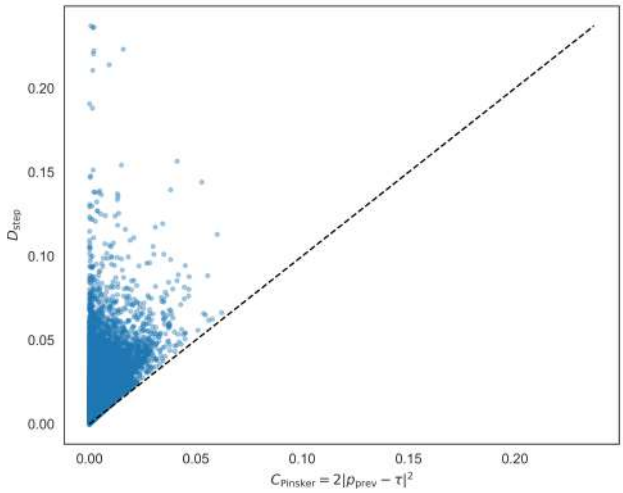
(c) XGBoost



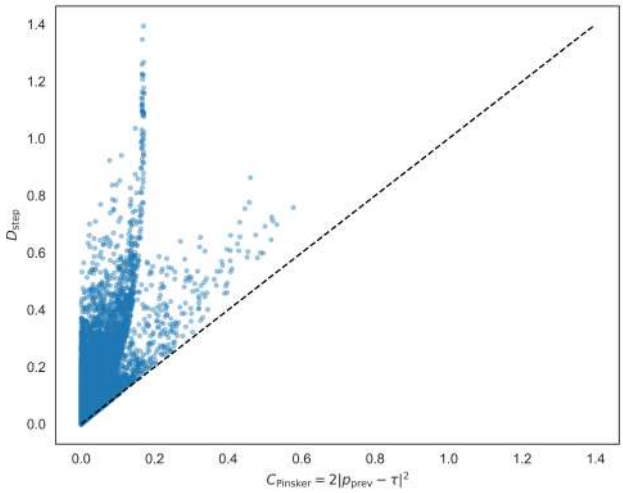
(d) ANN

Figure S3: Threshold-certificate scatter diagnostics for the PDAC biomarker dataset at $\tau = 0.7$.

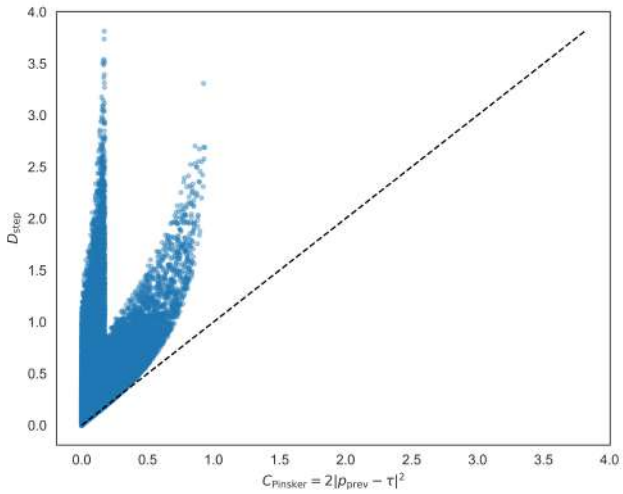
1.2 Credit card default



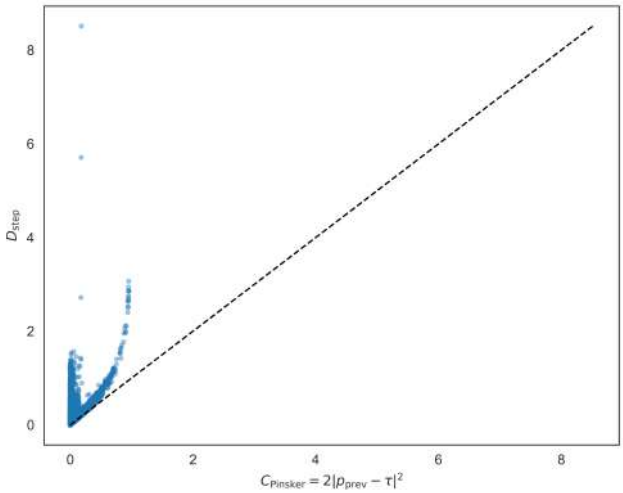
(a) Logistic regression



(b) Random forest

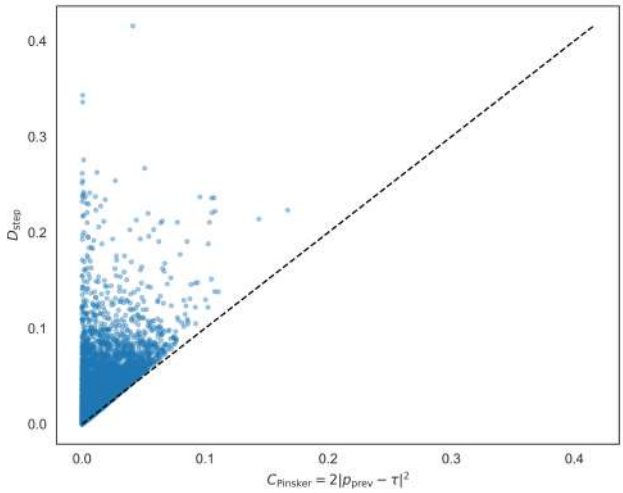


(c) XGBoost

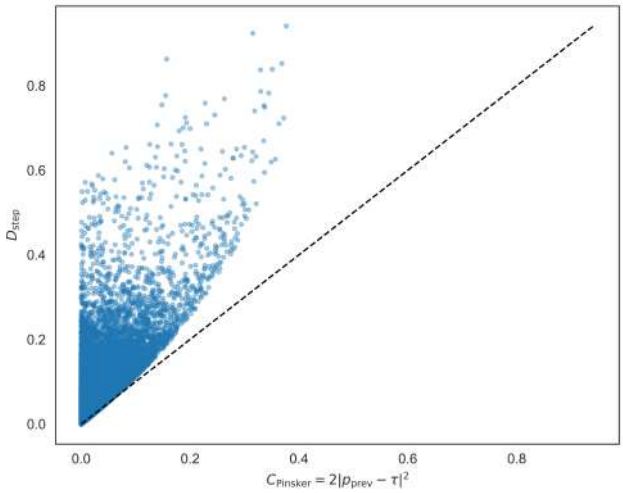


(d) ANN

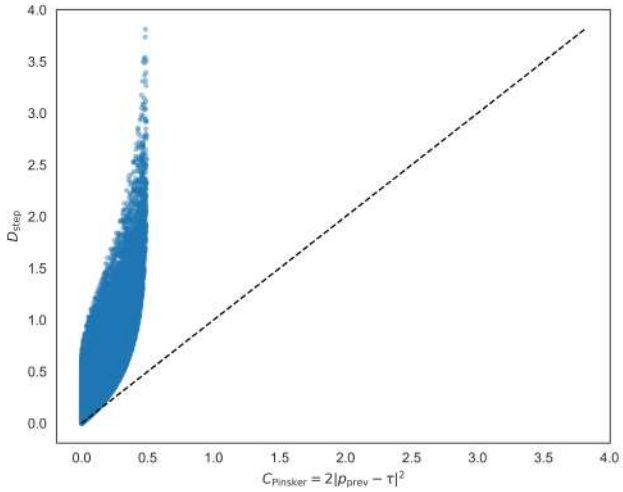
Figure S4: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.3$.



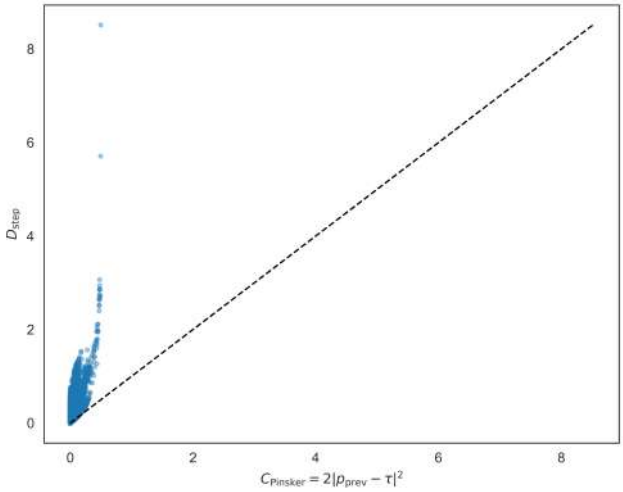
(a) Logistic regression



(b) Random forest

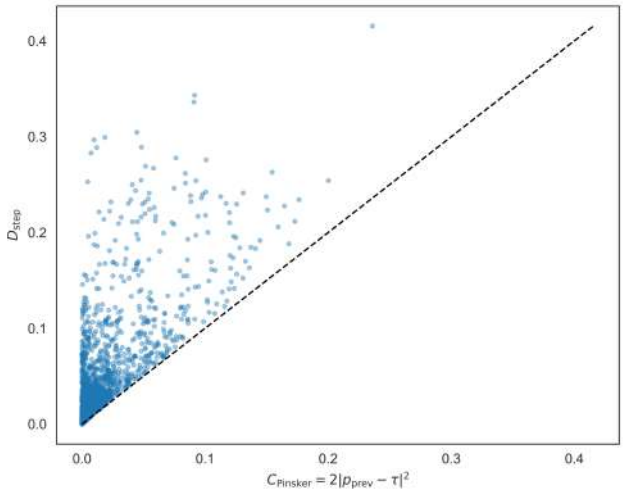


(c) XGBoost

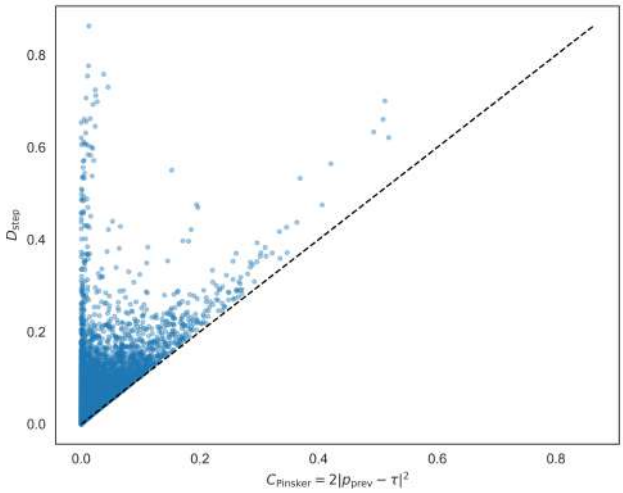


(d) ANN

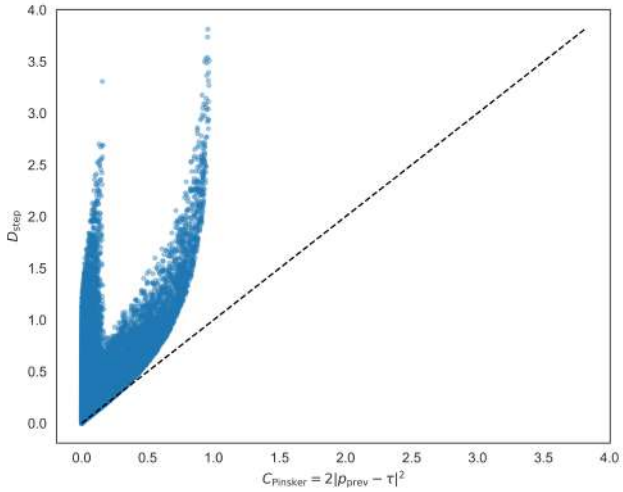
Figure S5: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.5$.



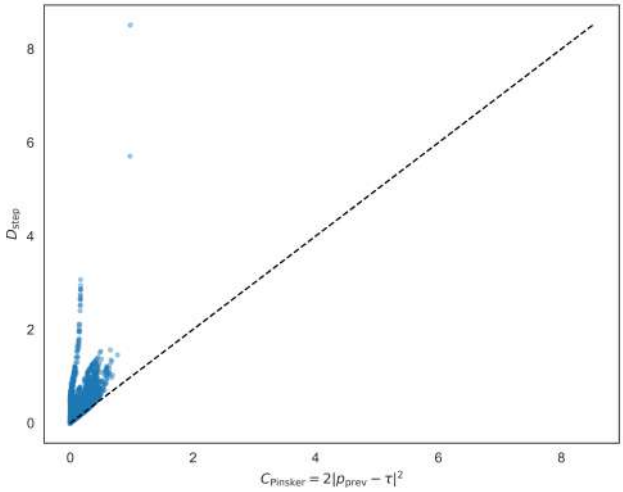
(a) Logistic regression



(b) Random forest



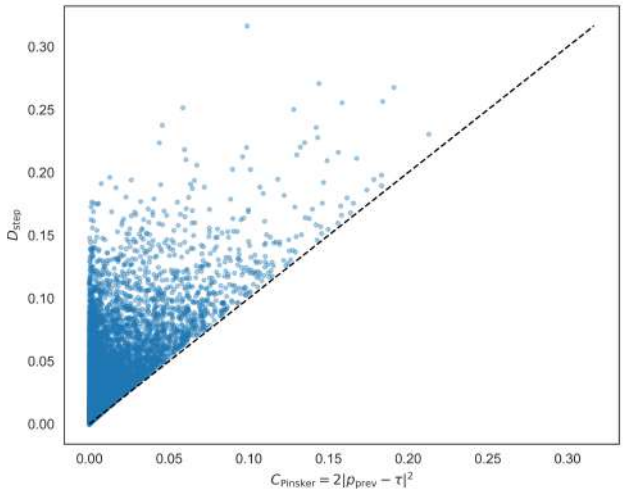
(c) XGBoost



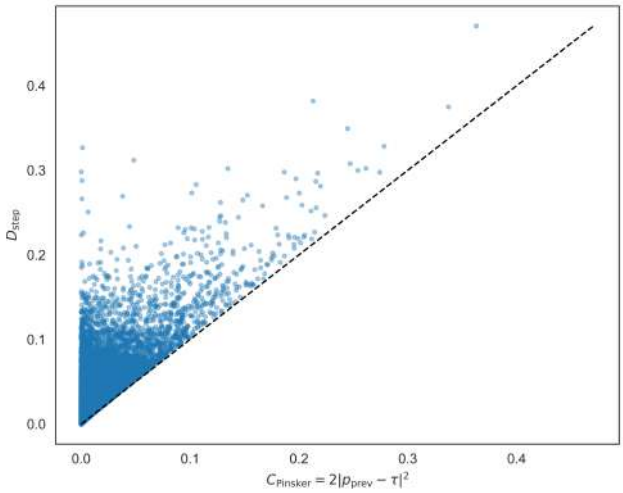
(d) ANN

Figure S6: Threshold-certificate scatter diagnostics for the credit-card-default dataset at $\tau = 0.7$.

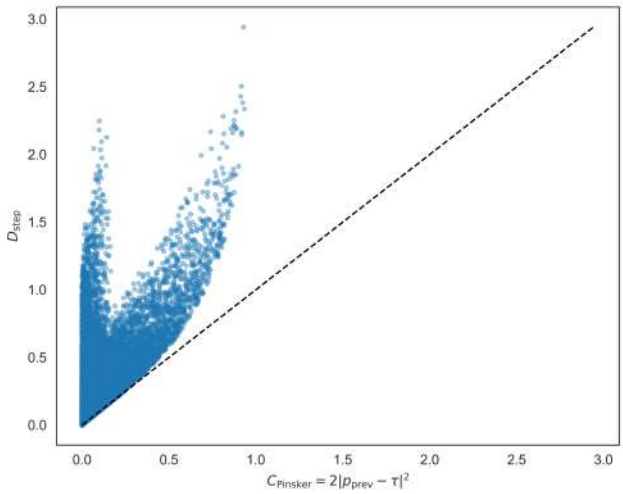
1.3 Electrical grid stability



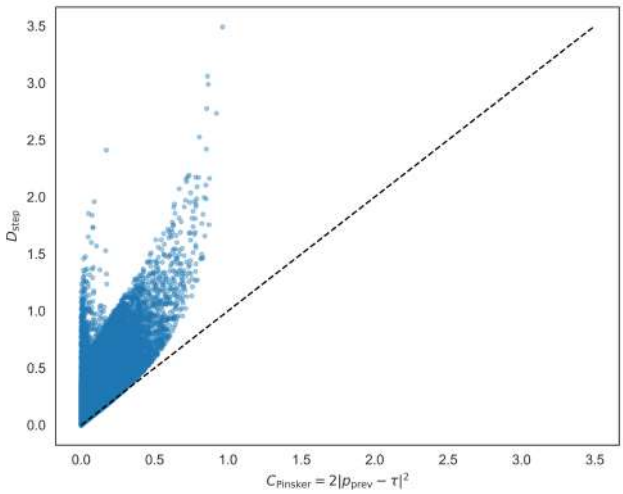
(a) Logistic regression



(b) Random forest

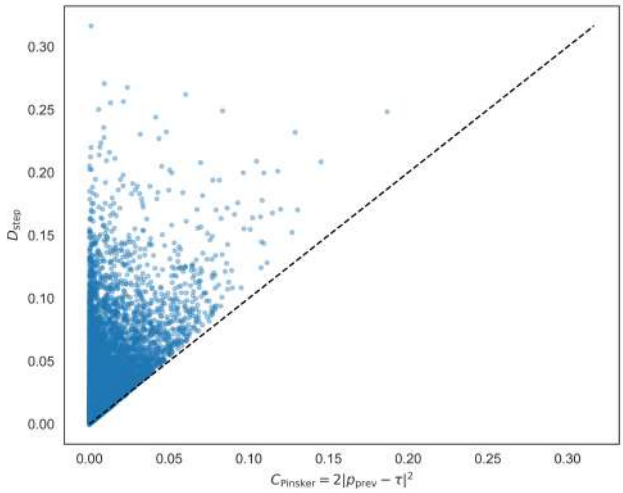


(c) XGBoost

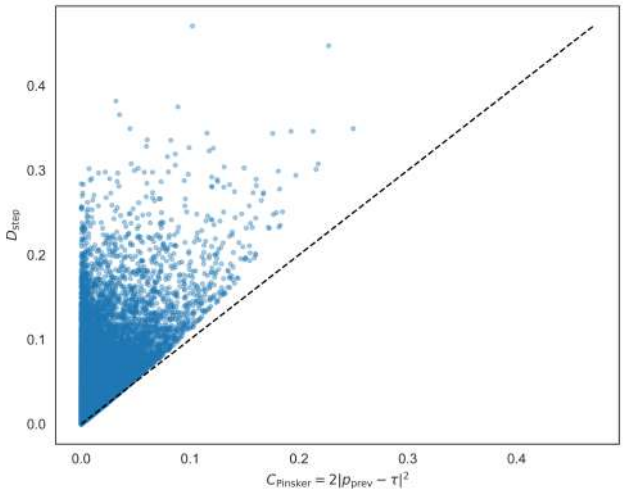


(d) ANN

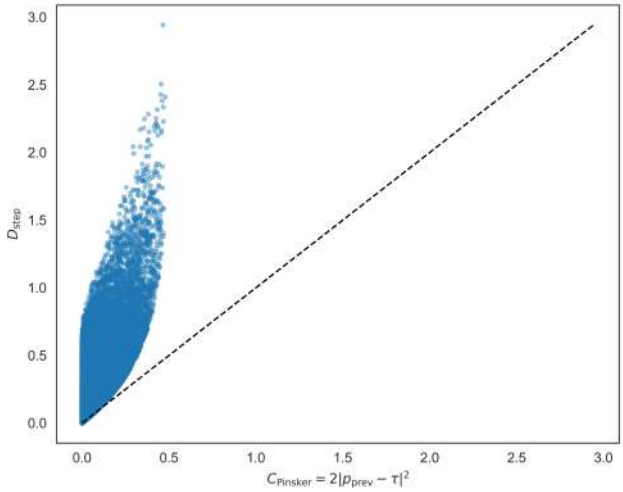
Figure S7: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.3$.



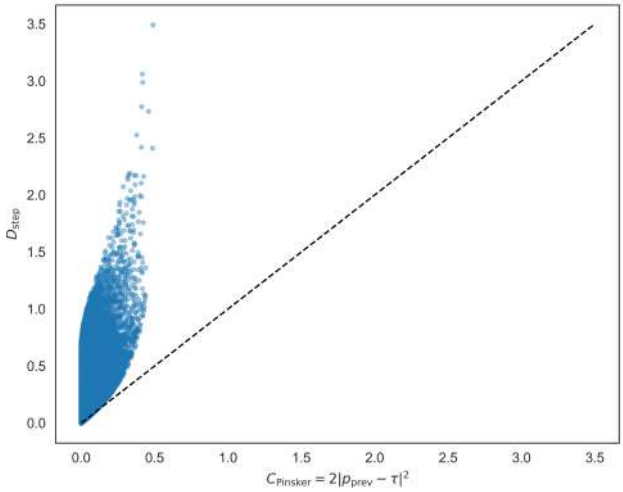
(a) Logistic regression



(b) Random forest

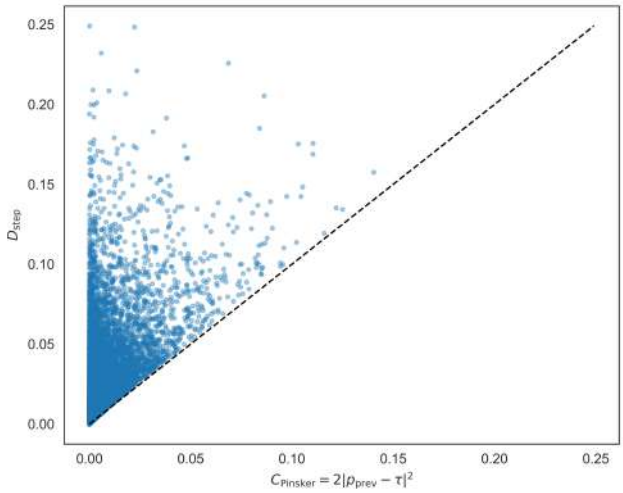


(c) XGBoost

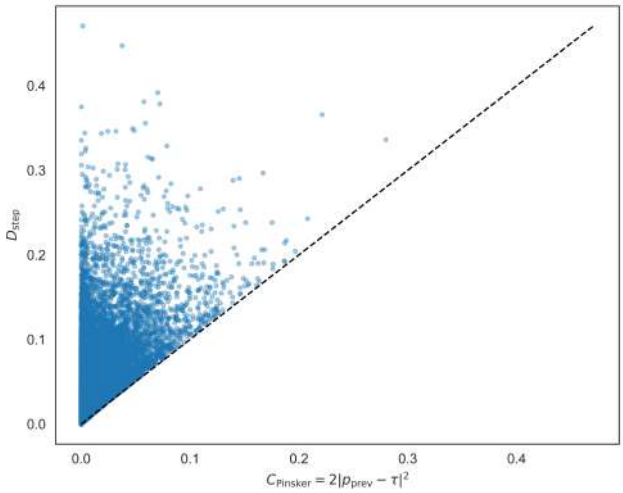


(d) ANN

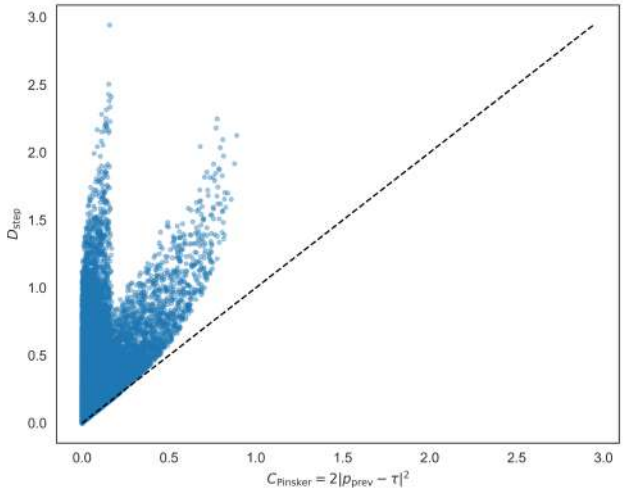
Figure S8: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.5$.



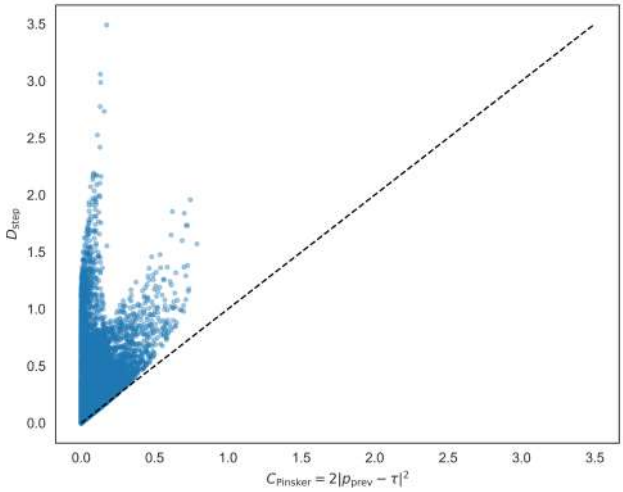
(a) Logistic regression



(b) Random forest



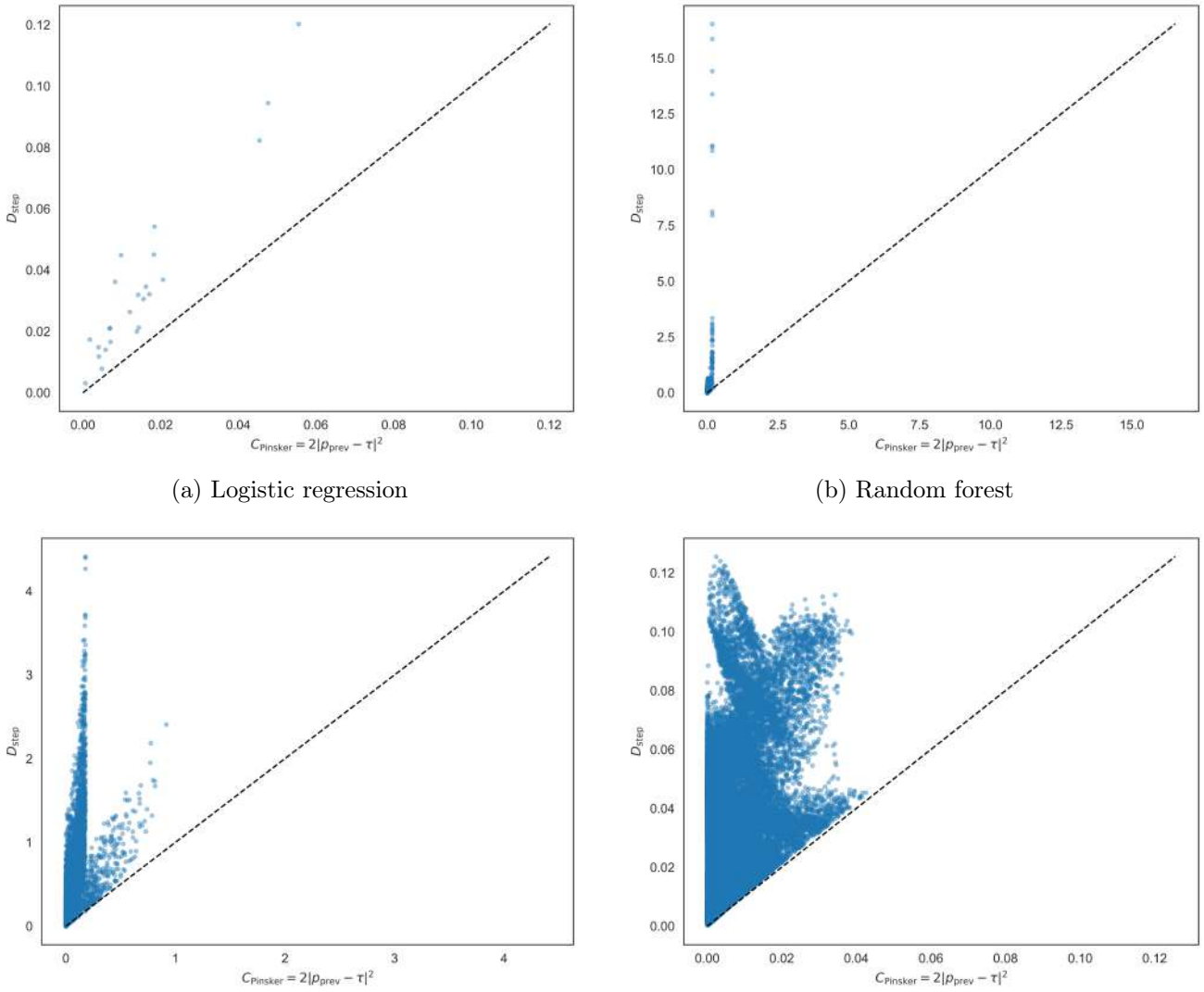
(c) XGBoost



(d) ANN

Figure S9: Threshold-certificate scatter diagnostics for the electrical-grid-stability dataset at $\tau = 0.7$.

1.4 Machine failure



(a) Logistic regression

(b) Random forest

(c) XGBoost

(d) ANN

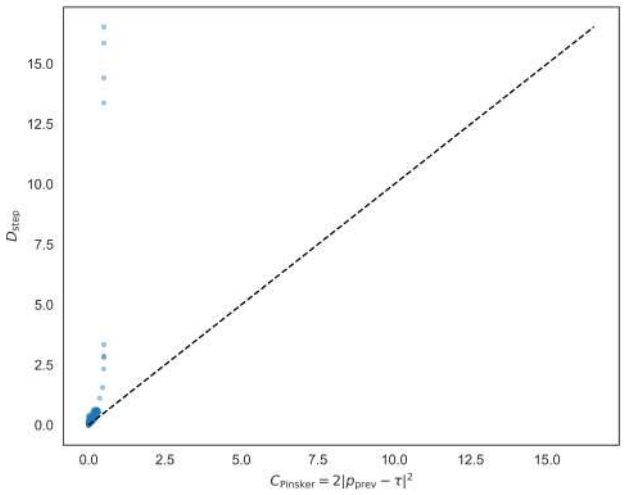
Figure S10: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.3$.

No flipped points were observed for

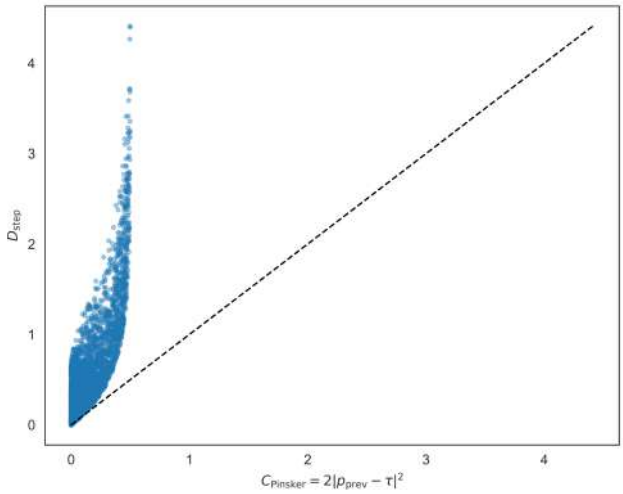
Logistic regression.

The Python script therefore did not generate a threshold-certificate scatter plot for this combination.

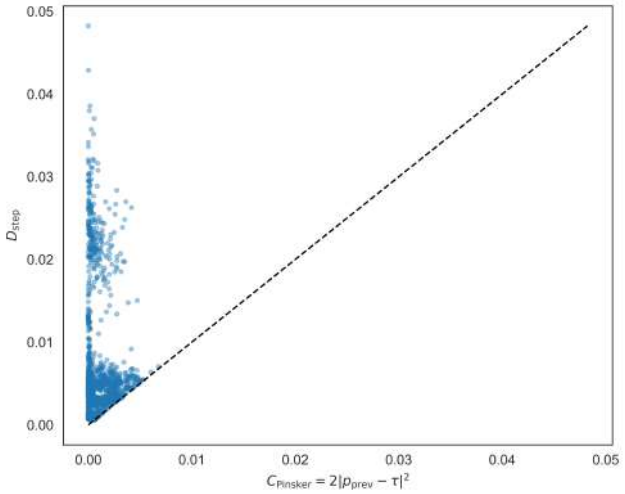
(a) Logistic regression



(b) Random forest



(c) XGBoost



(d) ANN

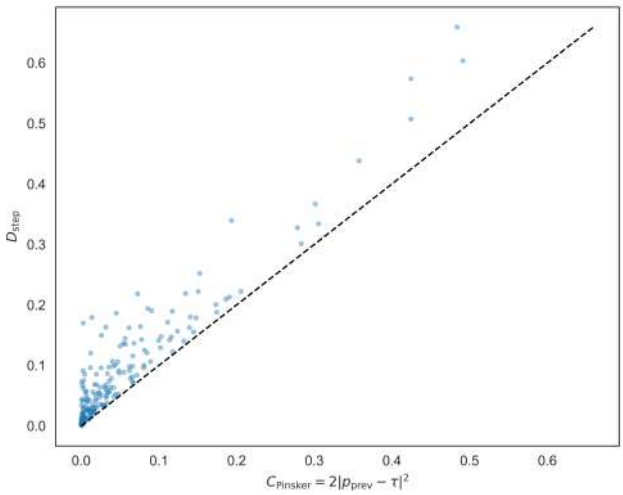
Figure S11: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.5$. Logistic regression produced no flipped points in this setting, so no scatter plot was generated for that panel.

No flipped points were observed for

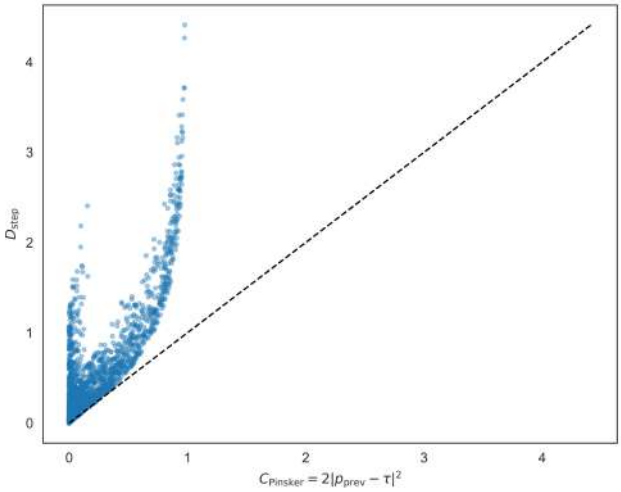
Logistic regression.

The Python script therefore did not generate a threshold-certificate scatter plot for this combination.

(a) Logistic regression



(b) Random forest



(c) XGBoost

No flipped points were observed for

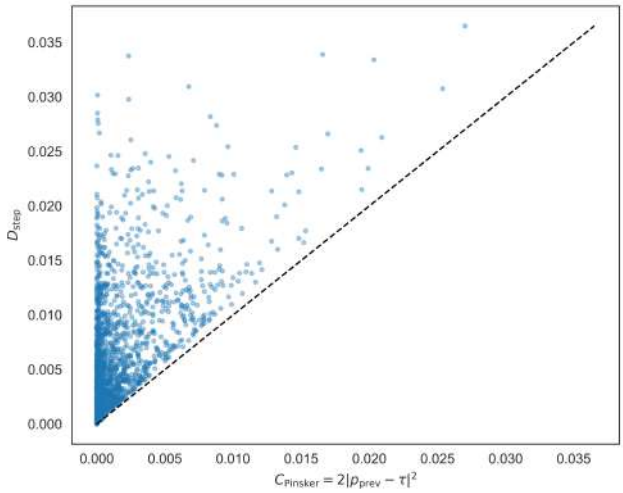
ANN.

The Python script therefore did not generate a threshold-certificate scatter plot for this combination.

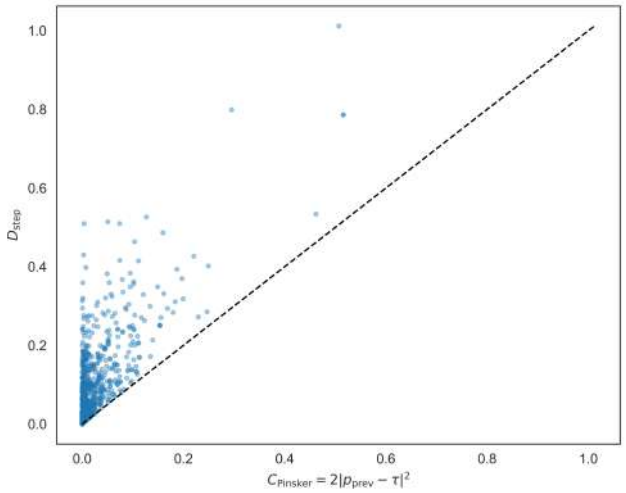
(d) ANN

Figure S12: Threshold-certificate scatter diagnostics for the machine-failure dataset at $\tau = 0.7$. Logistic regression and ANN produced no flipped points in this setting, so no scatter plots were generated for those panels.

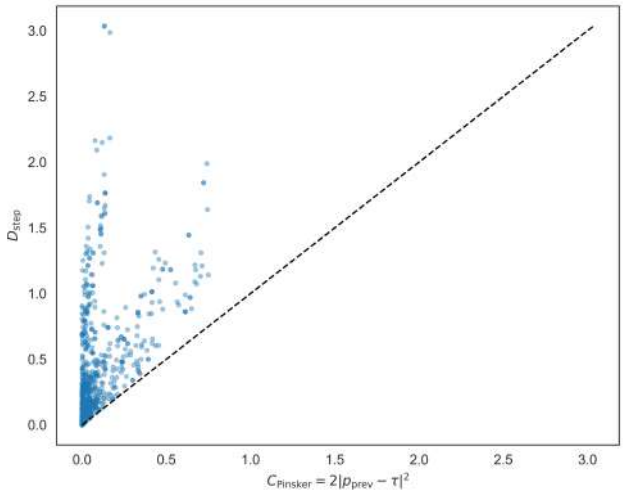
1.5 Banknote authentication



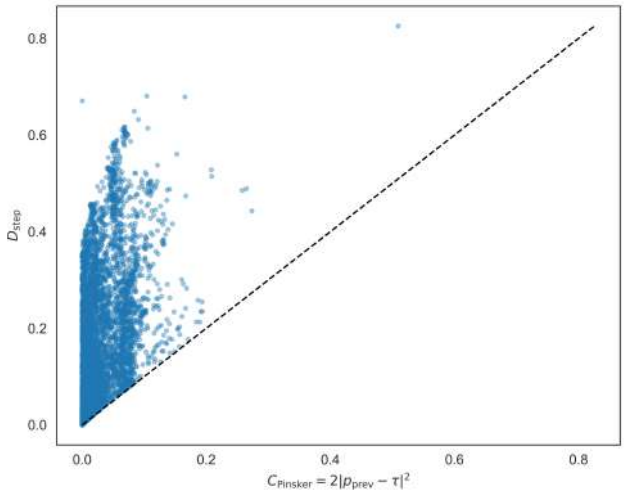
(a) Logistic regression



(b) Random forest

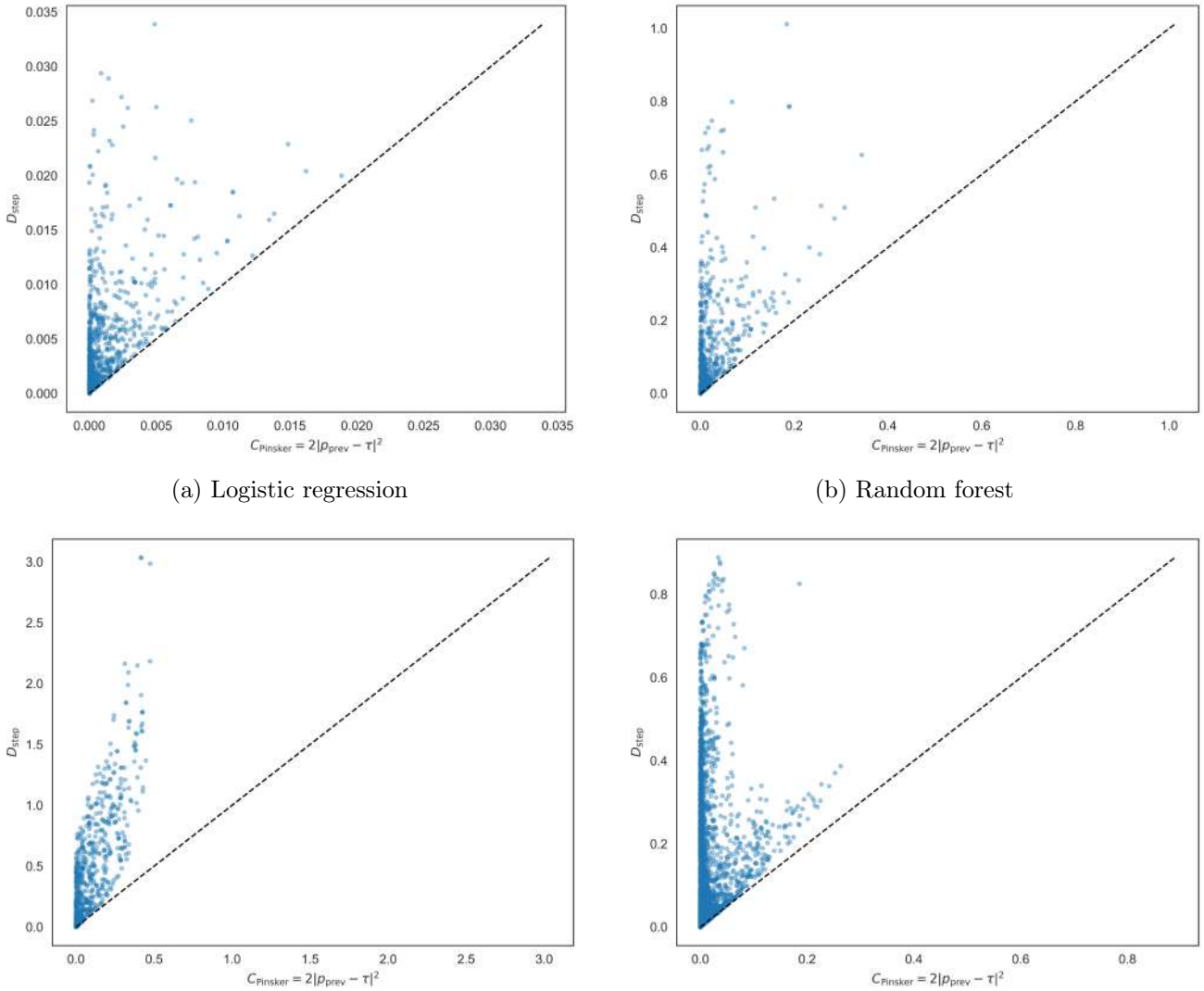


(c) XGBoost



(d) ANN

Figure S13: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.3$.



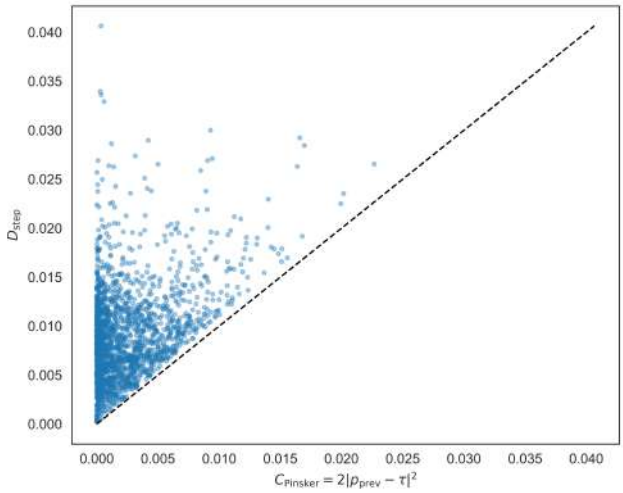
(a) Logistic regression

(b) Random forest

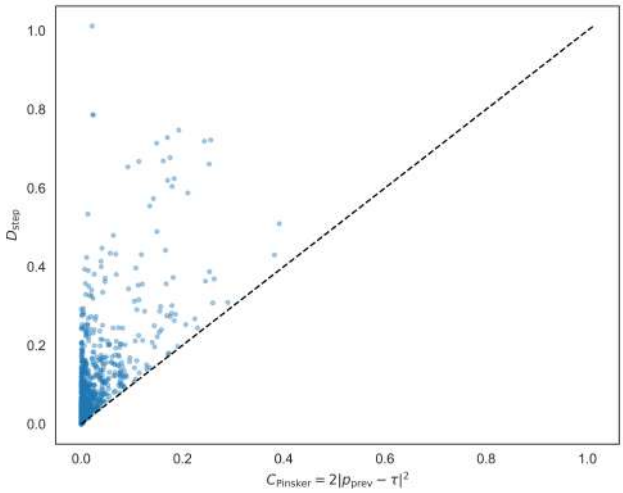
(c) XGBoost

(d) ANN

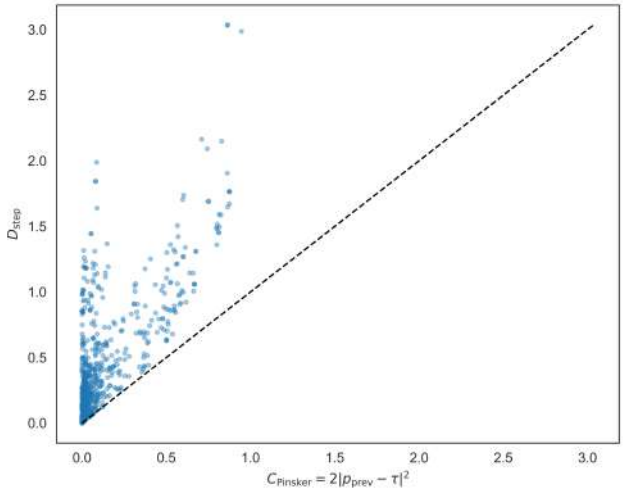
Figure S14: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.5$.



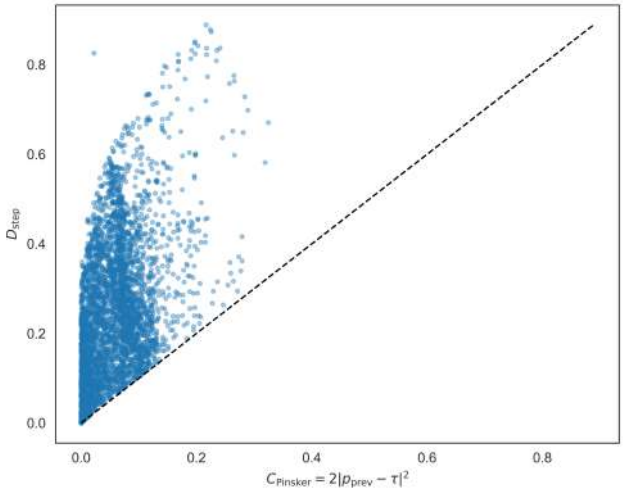
(a) Logistic regression



(b) Random forest



(c) XGBoost

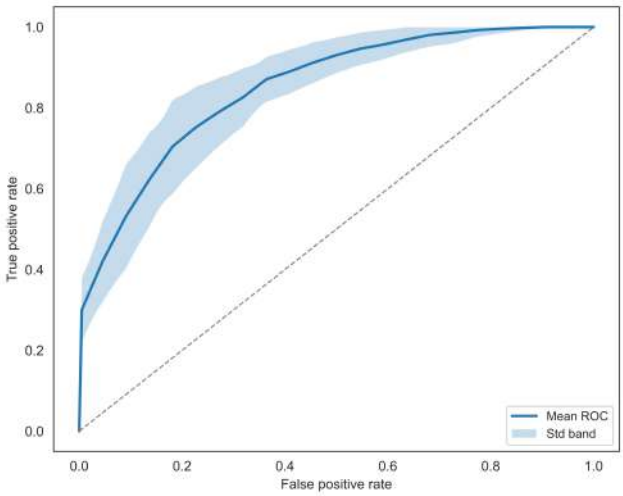


(d) ANN

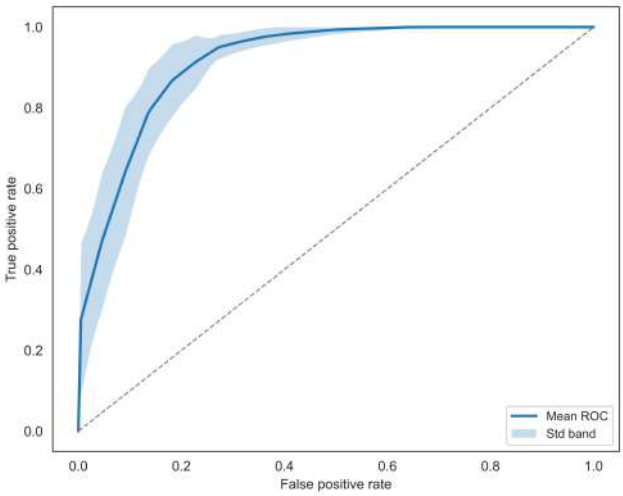
Figure S15: Threshold-certificate scatter diagnostics for the banknote-authentication dataset at $\tau = 0.7$.

2 Final ROC curves

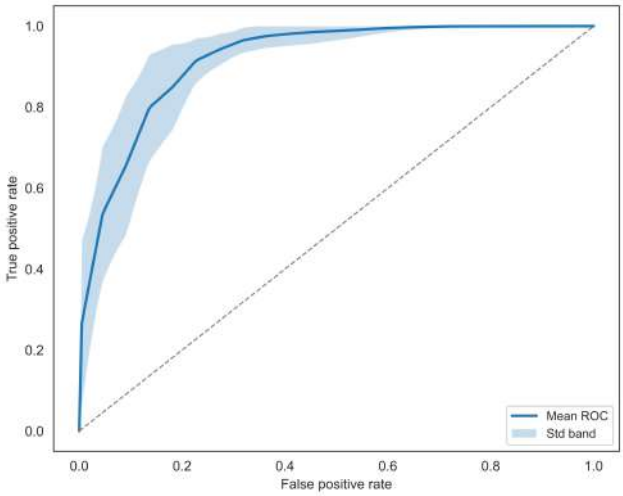
The following figures show the ROC curves at the largest training size for each dataset–model pair. The mean curve and ±1 standard-deviation band are computed across the 20 repeated holdout repetitions.



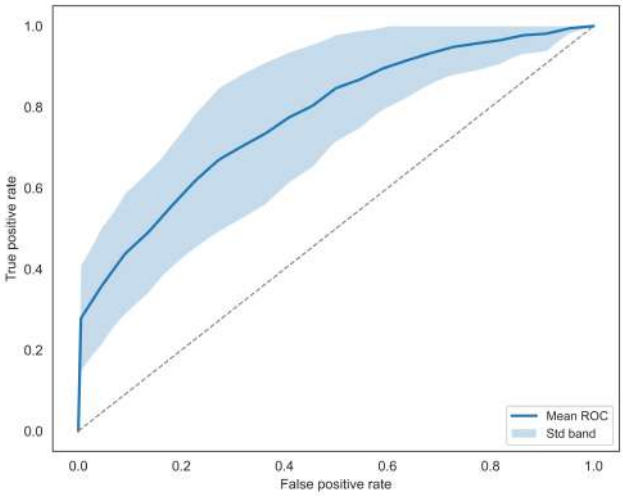
(a) Logistic regression



(b) Random forest

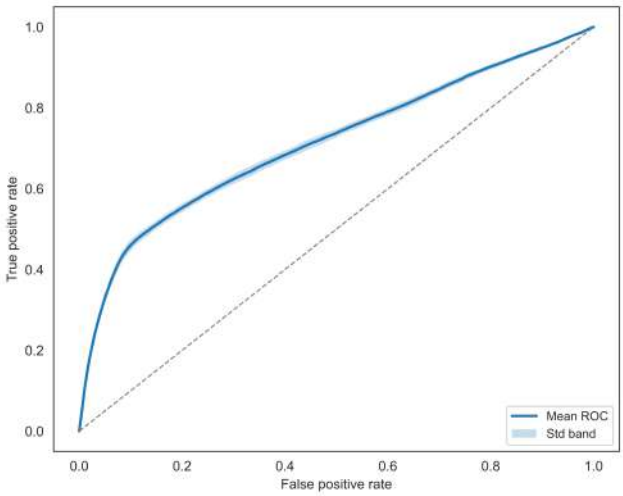


(c) XGBoost

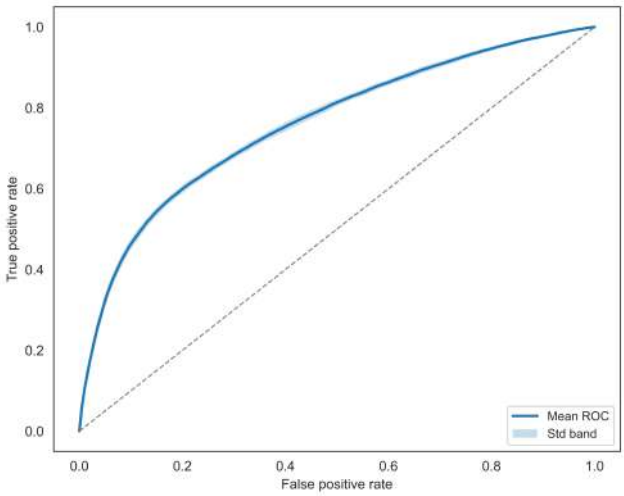


(d) ANN

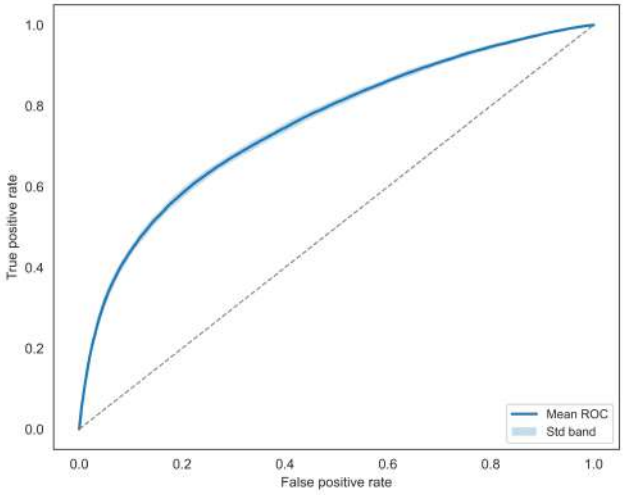
Figure S16: Final ROC curves for the PDAC biomarker dataset.



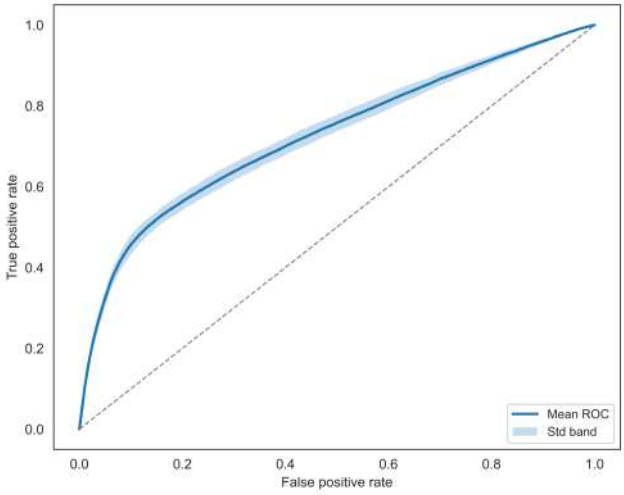
(a) Logistic regression



(b) Random forest

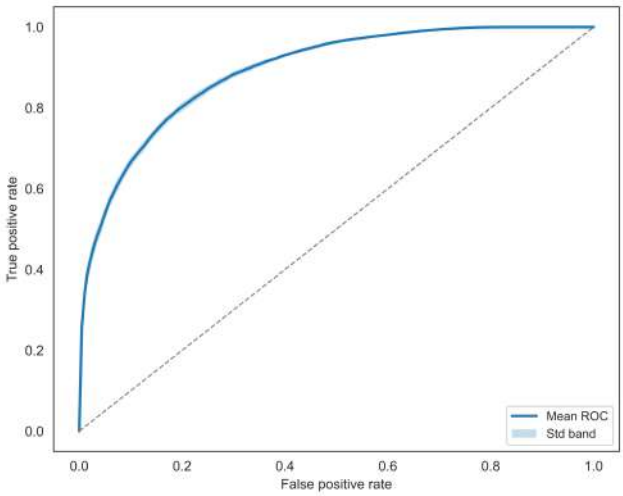


(c) XGBoost

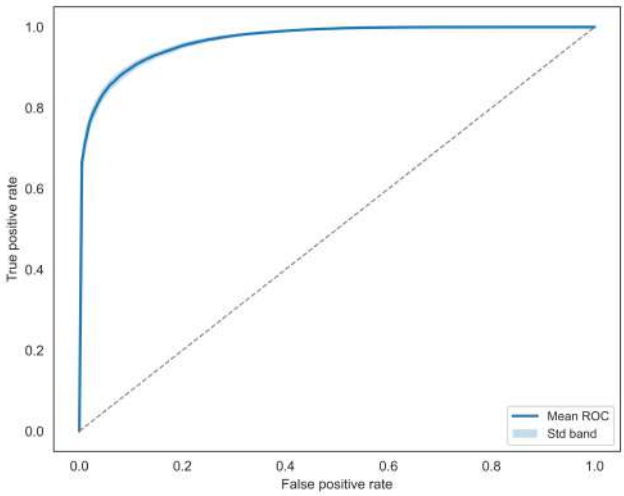


(d) ANN

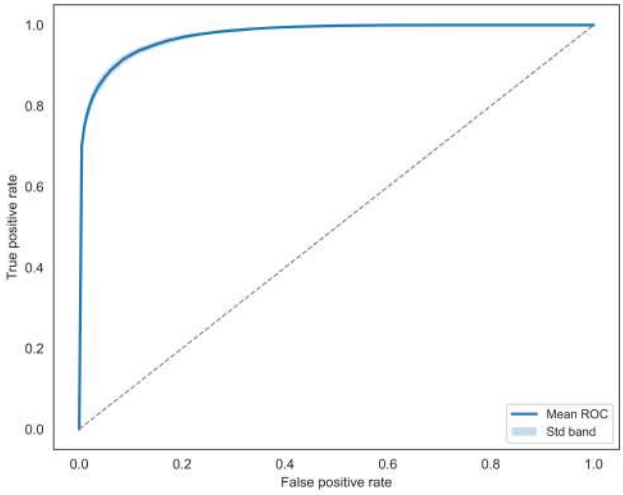
Figure S17: Final ROC curves for the credit-card-default dataset.



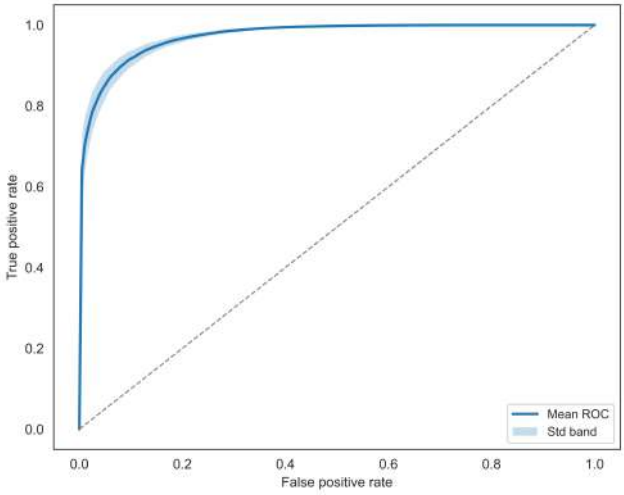
(a) Logistic regression



(b) Random forest

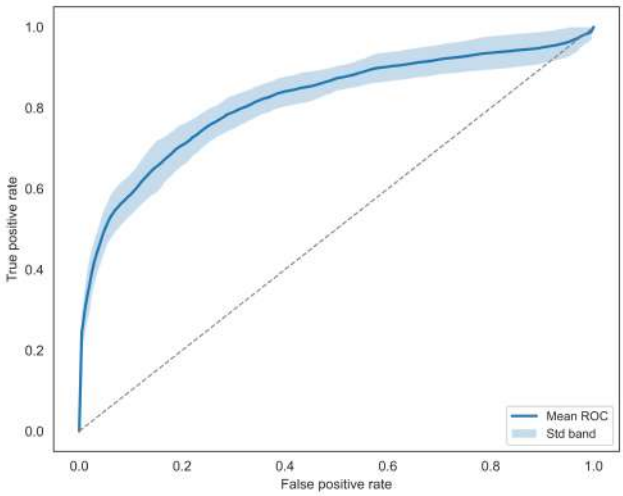


(c) XGBoost

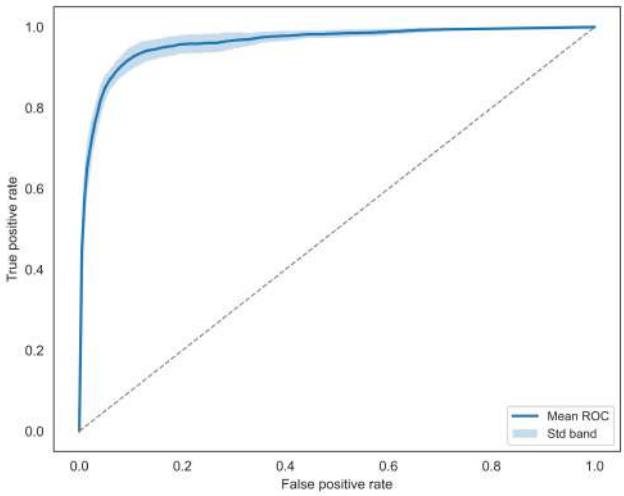


(d) ANN

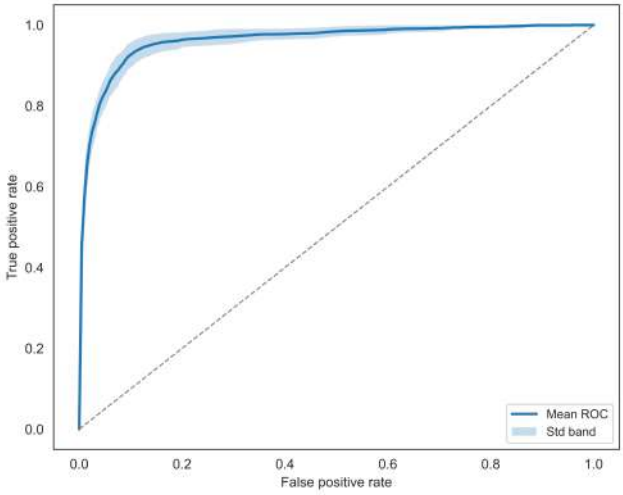
Figure S18: Final ROC curves for the electrical-grid-stability dataset.



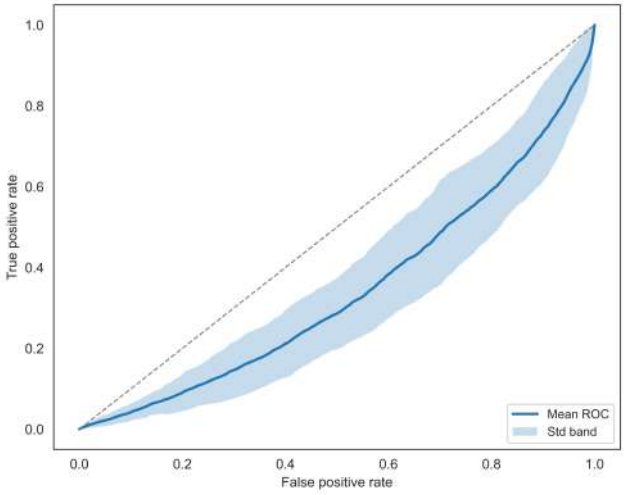
(a) Logistic regression



(b) Random forest

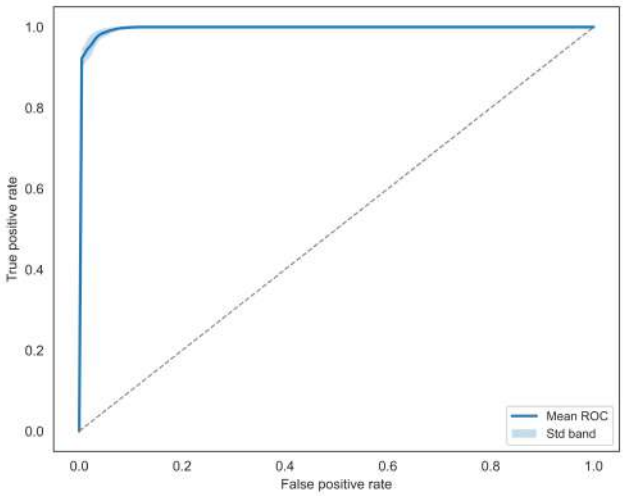


(c) XGBoost

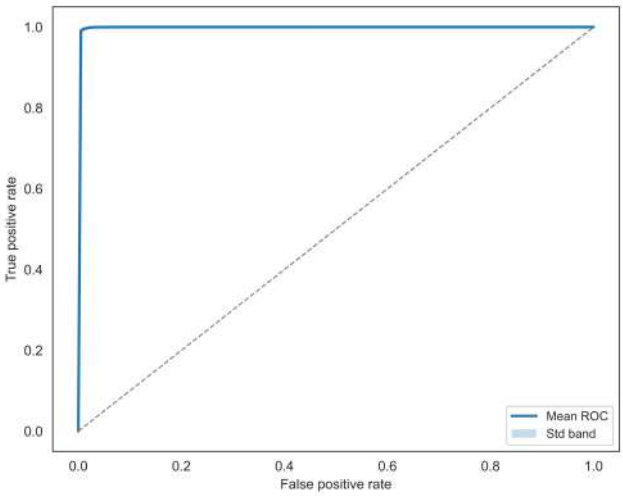


(d) ANN

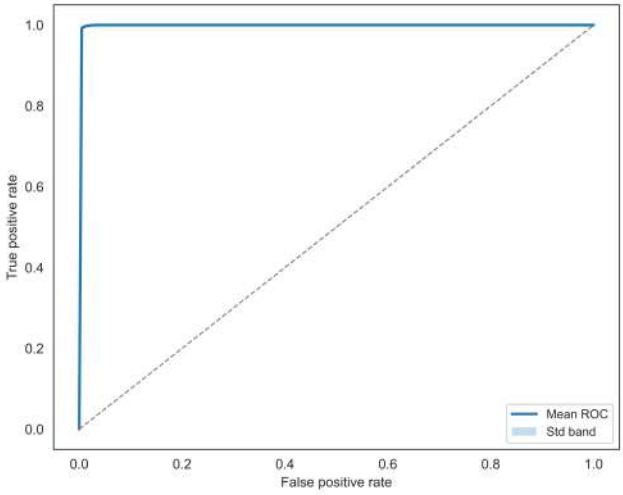
Figure S19: Final ROC curves for the machine-failure dataset.



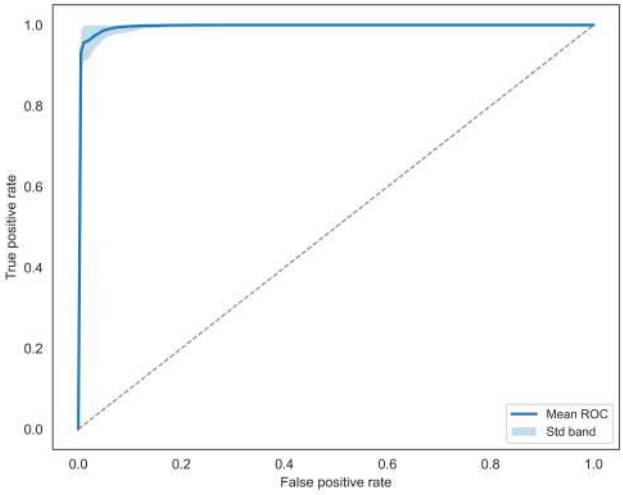
(a) Logistic regression



(b) Random forest



(c) XGBoost



(d) ANN

Figure S20: Final ROC curves for the banknote-authentication dataset.